Basic Concepts in Chemistry - Questions

1. Atoms

1. Give the symbol for each of the following atoms:

hydrogen	lithium	boron	nitrogen	fluorine	sodium
aluminium	phosphorus	chlorine	potassium	iron	silver
zinc	tin	bromine	barium	manganese	nickel

2. Give the names of the elements with the following symbols

U	Co	Cr
I	Pb	Hg
Au	Cu	Ca
Ar	S	Si
Mg	Ne	O
C	Be	He

3. Use your periodic table to answer the following questions.
   a) How many protons are there in magnesium atoms and in copper atoms?
   b) How many electrons are there in neutral atoms of sulfur and in neutral atoms of iodine?

4. a) Six atoms whose names begin with the letter "C" are carbon, chlorine, calcium, cobalt , copper and chromium. Give the symbol for each of these atoms.

   b) Four atoms whose symbols begin with the letter "N" are N, Na, Ne and Ni. Give the name of each of these atoms.

5. The symbols of some atoms are based on their Latin name. Complete the following table by giving the name and/or symbol for each atom:

English name	Latin name	Chemical symbol
	ferrum	
	natrium	
mercury	hydrargyrum	
	kalium	
lead	plumbum	
tin	stannum	
	auram	
silver	argentum	

6. Give the name and symbol of each of the following atoms:
   a) An atom that has an atomic number of 47
   b) An atom that has four more protons than carbon
   c) An atom with 29 protons, 34 neutrons and 29 electrons
   d) A neutral atom that has 20 electrons

7. Identify the following statements as true or false. If the statement is false, rewrite it to make it correct.

   a) The symbol of silicon is S and the symbol of sulfur is Su.

   b) An atom with 13 protons and 13 neutrons and a second atom with 13 protons and 15 neutrons are both atoms of aluminium.

   c) Zinc atoms contain one more proton than copper atoms.

   d) Protons have a charge of -1.

   e) Neutrons are negatively charged.

   f) Protons and neutrons are found in the nucleus of an atom.

   g) Electrons remain stationary at a constant distance from the nucleus.

   h) An electron has a charge of -1.

   i) An atom with 5 protons and 6 electrons would be a neutral atom.

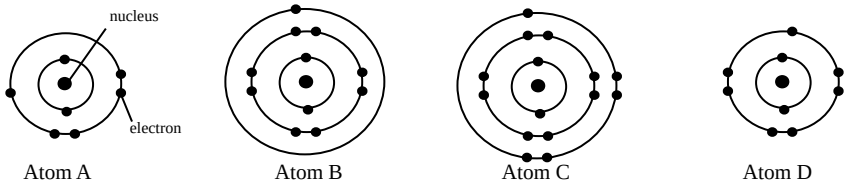
   j) If a neutral atom has an atomic number of 17, then it must contain 17 electrons.

2. Electron Arrangement in Atoms

8. Draw diagrams to show the electron arrangement for the following atoms:

   a) lithium b) sodium c) an atom with an atomic number of 17.

9. To answer the questions below, refer to the <u>neutral</u> atoms A to D shown in the following diagrams :



   a) Which atom has an electron configuration of 2, 8, 1?

   b) Which atom has 5 protons in its nucleus?

   c) Which atom contains the largest number of protons in its nucleus?

   d) Which atom has an atomic number of 9?

   e) Which atom is a sodium atom?

   f) Which atoms have only 5 electrons in their outermost shell?

   g) Which atoms have two electrons in the first shell?

10. Give the electron configuration (e.g. 2, 6) of the following atoms

   a) hydrogen b) magnesium c) an atom with an atomic number of 7.

11. All the following statements concerning neutral atoms are incorrect. Rewrite them to make them correct.

a) An atom of aluminium has an electron configuration of 8, 5.

b) An atom with an atomic number of 20 would have an electron configuration of 2, 8, 10

c) An atom of phosphorus has 8 electrons in its first shell.

d) A maximum number of 10 electrons can exist in the second shell of an atom.

e) The first electron shell is further from the nucleus than the second shell.

f) Argon has eight electrons in the second shell and potassium has nine electrons in the second shell.

g) Lithium has 3 electrons in the first electron shell.

h) Compared to helium, beryllium has two more electrons in the first shell.

i) Carbon has 6 electrons in its second shell.

j) An atom with an atomic number of 11 would have 11 electrons in its third shell.

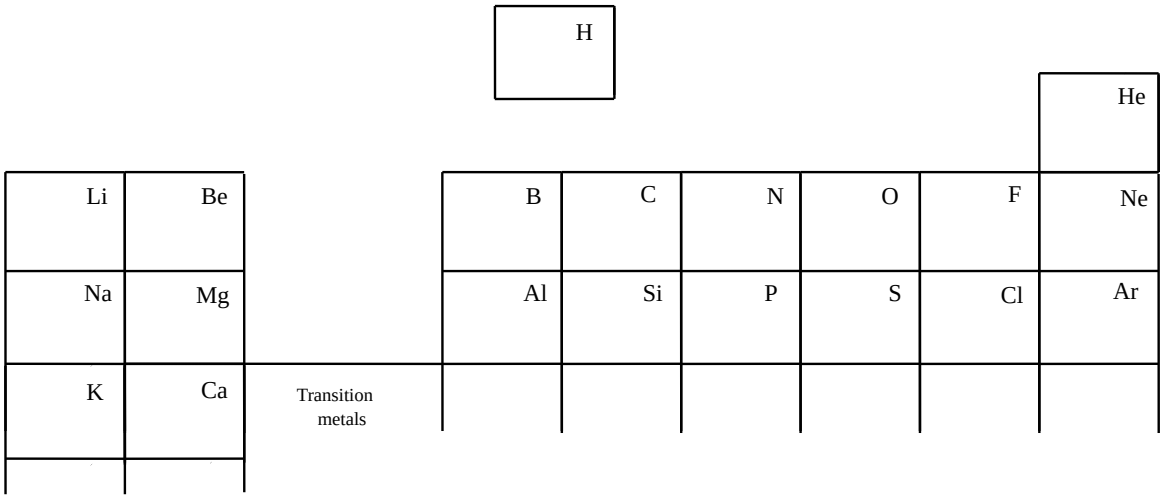
3. The Periodic Table

12. On the periodic table below:

a) write the atomic number for each element

a) write the electron configuration for each of the elements from hydrogen to calcium

b) write the group numbers at the top of each group.



13. Use a periodic table to classify each of the following elements as a metal or a non-metal:

lithium	carbon	phosphorus	chlorine	potassium	calcium
nitrogen	oxygen	fluorine	zinc	bromine	iodine
sodium	magnesium	sulfur	barium	hydrogen	copper

14. Give the name and symbol of an element/atom for each of the following descriptions:

a) An element in Group 2 that has electrons in only the first two electron shells

b) A Group 4 element that is a non-metal

c) An atom with an electron configuration of 2, 8, 6

d) An element with similar properties as fluorine.

e) An element in the same group as neon, but with fewer protons than neon.

f) An element with one more proton than argon

14. (cont) g) A neutral atom with only two electrons in its first shell and seven electrons in its second shell.

h) A neutral atom with an atomic number of less than 10, which has only 6 electrons in its outermost electron shell.

15.

chlorine	carbon	magnesium	copper	argon

Choose from the elements written in the box, to answer the questions below:

a) An element with an atomic number of 17.

b) An element in Group 4.

c) An element with similar properties as calcium.

d) The element with the largest number of protons.

e) The metal elements.

f) A non-metal element from Group 7.

g) An element with only 2 electrons in its outermost shell.

h) An element whose neutral atoms would have 12 electrons.